Articaine and epinephrine: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Articaine and epinephrine: Patient drug information” and ”Articaine and epinephrine: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Articadent;
- Orabloc;
- Septocaine with Epinephrine 1:100,000;
- Septocaine with Epinephrine 1:200,000;
- Zorcaine

Brand Names: Canada

- Astracaine with Epinephrine 1:200,000;
- Astracaine with Epinephrine forte 1:100,000;
- Karticaine;
- Karticaine Forte;
- Orabloc 1:100,000;
- Orabloc 1:200,000;
- Posicaine N;
- Posicaine SP;
- Septanest N;
- Septanest SP;
- Ultracaine DS;
- Ultracaine DS Forte;
- Zorcaine

Pharmacologic Category

- Local Anesthetic

Dosing: Adult

Dental anesthesia

Dental anesthesia: Submucosal infiltration and/or nerve block: Articaine 4%/epinephrine: **Note:** These dosages are guides only; other dosages may be used; however, do not exceed maximum recommended dose. The actual volumes to be used depend upon a number of factors, such as type and extent of surgical procedure, depth of anesthesia, degree of muscular relaxation, and condition of the patient. In all cases, the smallest dose that will produce the desired result should be given.

For most routine dental procedures, epinephrine 1:200,000 is preferred; when more pronounced hemostasis or improved visualization of the surgical field are required, epinephrine 1:100,000 may be used. Dosages should be reduced for patients with cardiac disease and acutely ill and/or debilitated patients:

Infiltration: 0.5 to 2.5 mL; total dose of articaine: 20 to 100 mg; maximum dose of articaine: 7 mg/kg (0.175 mL/kg).

Nerve block: 0.5 to 3.4 mL; total dose of articaine: 20 to 136 mg; maximum dose of articaine: 7 mg/kg (0.175 mL/kg).

Oral surgery: 1 to 5.1 mL; total dose of articaine: 40 to 204 mg; maximum dose of articaine: 7 mg/kg (0.175 mL/kg).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion of active drug in the urine.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied). Use with caution in patients with severe hepatic disease.

Dosing: Older Adult

Dental anesthesia: Submucosal infiltration and/or nerve block: Articaine 4%/epinephrine: **Note:** These dosages are guides only; other dosages may be used; however, do not exceed maximum recommended dose. The actual volumes to be used depend upon a number of factors, such as type and extent of surgical procedure, depth of anesthesia, degree of muscular relaxation, and condition of the patient. In all cases, the smallest dose that will produce the desired result should be given. For most routine dental procedures, epinephrine 1:200,000 is preferred; when more pronounced hemostasis or improved visualization of the surgical field are required, epinephrine 1:100,000 may be used. Dosages should be reduced for patients with cardiac disease and acutely ill and/or debilitated patients:

65 to 75 years:

Simple procedures: 0.43 to 4.76 mg/kg of articaine.

Complex procedures: 1.05 to 4.27 mg/kg of articaine.

≥75 years:

Simple procedures: 0.78 to 4.76 mg/kg of articaine.

Complex procedures: 1.12 to 2.17 mg/kg of articaine.

Dosing: Pediatric

(For additional information see "Articaine and epinephrine: Pediatric drug information")

Dental anesthesia

Dental anesthesia: Note: The provided dosages are guides only; other dosages may be necessary; however, do not exceed maximum recommended dose. The actual volumes to be used depend upon a number of factors, such as type and extent of surgical procedure, depth of anesthesia, degree of muscular relaxation, and condition of the patient. In all cases, the smallest dose that will produce the desired result should be used. Two concentrations of epinephrine (1:100,000 or 1:200,00) with 4% articaine are available; when more pronounced hemostasis or improved visualization of the surgical field are required, epinephrine 1:100,000 may be used; in clinical trials of pediatric patients 4 to 16 years of age, the 1:100,000 was also used; in adults, the manufacturer recommends epinephrine 1:200,000 for most routine dental procedures. Dosages should be reduced for patients with cardiac disease and acutely ill and/or debilitated patients. Dosing presented in variable unit (mg/kg, mg, mL/kg, and mL); use extra precaution to verify dosing units.

Children ≥4 years and Adolescents ≤16 years: Submucosal infiltration and/or nerve block: Articaine 4%/epinephrine: Injection:

Simple procedures: Reported range: 0.76 to 5.65 mg/kg of articaine; maximum articaine dose: 7 mg/kg (0.175 mL/kg of 4% solution)

Complex procedures: 0.37 to 7 mg/kg of articaine; maximum articaine dose: 7 mg/kg (0.175 mL/kg of 4% solution)

Adolescents ≥17 years: Submucosal infiltration and/or nerve block: Articaine 4%/epinephrine: Injection:

Infiltration: 0.5 to 2.5 mL (total articaine dose: 20 to 100 mg); not to exceed 7 mg/kg (0.175 mL/kg of 4% solution) of articaine

Nerve block: 0.5 to 3.4 mL (total articaine dose: 20 to 136 mg); not to exceed 7 mg/kg (0.175 mL/kg of 4% solution) of articaine

Oral surgery: 1 to 5.1 mL (total articaine dose: 40 to 204 mg); not to exceed 7 mg/kg (0.175 mL/kg of 4% solution) of articaine

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied).

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied). Use with caution in patients with severe hepatic disease.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Nervous system: Pain (6% to 13%)

1% to 10%:

Cardiovascular: Palpitations (1%)

Gastrointestinal: Gingivitis (1%), nausea and vomiting (2%)

Hypersensitivity: Facial edema (1%)

Infection: Infection (1%)

Nervous system: Drowsiness (≤1%), headache (3% to 5%), numbness (≤1%), paresthesia (1%), tingling sensation (≤1%), trismus (≤2%)

Otic: Otalgia (≤1%)

Respiratory: Cough (1%)

Miscellaneous: Swelling (2% to 3%)

<1%:

Cardiovascular: Edema, increased blood pressure, syncope, tachycardia

Dermatologic: Ecchymoses, pruritus

Endocrine & metabolic: Increased thirst

Gastrointestinal: Dysgeusia, dyspepsia, gingival hemorrhage, glossitis, oral mucosa ulcer, sialorrhea, stomatitis, xerostomia

Hematologic & oncologic: Hemorrhage, lymphadenopathy

Hypersensitivity: Tongue edema

Local: Burning sensation at injection site, pain at injection site

Nervous system: Asthenia, dizziness, facial nerve paralysis, hyperesthesia, malaise, migraine, nervousness, neuropathy

Neuromuscular & skeletal: Arthralgia, back pain, myalgia, neck pain, neuromuscular symptoms (exacerbation of Kearns-Sayre syndrome), osteomyelitis

Respiratory: Paranasal sinus congestion, pharyngitis, rhinitis, sinus pain

Postmarketing:

Cardiovascular: Ischemia

Miscellaneous: Tissue necrosis

Contraindications

Sulfite hypersensitivity.

Canadian labeling: Additional contraindications (not in US labeling): **Note:** May vary by product; also refer to product labeling: Hypersensitivity to articaine, epinephrine, or any component of the formulation; allergies to dental anesthetics; sepsis near the proposed injection site; severe shock; paroxysmal tachycardia; frequent arrhythmia; neurological disease; severe hypertension; patients with asthma who may have bronchospastic allergic reactions induced by sulfites; epilepsy (not controlled by treatment); intravascular use.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Local toxicity: Epinephrine may cause local toxicity, including ischemic injury or necrosis.

• Methemoglobinemia: Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months of age are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, lightheadedness, fatigue).

• Systemic toxicity: May occur. Systemic absorption of local anesthetics may produce cardiovascular and/or CNS effects. Toxic blood concentrations of local anesthetics depress cardiac conduction and excitability, which may lead to AV block, ventricular arrhythmias, and cardiac arrest (sometimes resulting in death). In addition, myocardial contractility is depressed and peripheral vasodilation occurs, leading to decreased cardiac output and arterial blood pressure. Restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression, or drowsiness may be early warning signs of CNS toxicity. Small doses of local anesthetics injected into dental blocks may produce adverse reactions similar to systemic toxicity, including confusion, convulsions, respiratory depression and/or respiratory arrest, and cardiovascular stimulation or depression; these reactions may be due to intra-arterial injection of the local anesthetic with retrograde flow to the cerebral circulation. Constantly monitor cardiovascular and respiratory vital signs and patient's state of consciousness carefully following each injection.

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with impaired cardiovascular function, including patients with heart block. Use local anesthetics containing a vasoconstrictor with caution in patients with vascular disease; patients with peripheral vascular disease or hypertensive vascular disease may exhibit exaggerated vasoconstrictor response, possibly resulting in ischemic injury or necrosis. Dosages should be reduced for patients with cardiac disease.

• Hepatic impairment: Use with caution in patients with severe hepatic disease (has not been studied).

Special populations:

• Acutely-ill/debilitated patients: Administer reduced dosages, commensurate with age and physical condition, to debilitated and/or acutely-ill patients.

• Older adult: Use with caution in the elderly; administer reduced dosages to commensurate with age and physical condition.

• Pediatric: Administer reduced dosages, commensurate with age and physical condition, to pediatric patients.

Dosage form specific issues:

• Sodium metabisulfite: May contain sodium metabisulfite, which may cause allergic-type reactions (including anaphylactic symptoms, and life-threatening or less severe asthmatic episodes) in certain susceptible patients. The overall prevalence of the sulfite sensitivity in the general population is unknown, and is seen more frequently in asthmatic than in nonasthmatic persons.

Other warnings/precautions:

• Administration: Avoid intravascular injection; accidental intravascular injection may be associated with convulsions, followed by CNS or cardiorespiratory depression and coma, progressing ultimately to respiratory arrest. Aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided.

• Appropriate use: To avoid serious adverse effects and high plasma levels, use the lowest dosage resulting in effective anesthesia. Repeated doses may cause significant increases in blood levels due to the possibility of accumulation of the drug or its metabolites. Dosage recommendations should not be exceeded.

• Trained personnel: Health care providers should be well trained in diagnosis and management of emergencies that may arise from the use of these agents. Resuscitative equipment, oxygen, and other resuscitative drugs should be available for immediate use.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Injection, solution [for dental use]:

Articadent: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:100,000 (1.7 mL) [contains sodium metabisulfite]

Orabloc: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:100,000 (1.8 mL) [contains sodium metabisulfite]

Orabloc: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:200,000 (1.8 mL) [contains sodium metabisulfite]

Septocaine with epinephrine 1:100,000: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:100,000 (1.7 mL) [contains sodium metabisulfite]

Septocaine with epinephrine 1:200,000: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:200,000 (1.7 mL) [contains sodium metabisulfite]

Zorcaine: Articaine hydrochloride 4% [40 mg/mL] and epinephrine 1:100,000 (1.7 mL) [contains sodium metabisulfite]

Generic Equivalent Available: US

No

Pricing: US

Solution Cartridge (Articadent Dental Injection)

4%-1:100000 (per mL): $0.66

Solution Cartridge (Orabloc Injection)

4%-1:100000 (per mL): $0.51

4%-1:200000 (per mL): $0.51

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Injection, solution [for dental use]:

Astracaine with epinephrine 1:200,000: Articaine hydrochloride 4% and epinephrine 1:200,000 (1.8 mL)

Astracaine Forte with epinephrine forte 1:100,000: Articaine hydrochloride 4% and epinephrine 1:100,000 (1.8 mL)

Septanest N: Articaine hydrochloride 4% and epinephrine 1:200,000 (1.7 mL)

Septanest SP: Articaine hydrochloride 4% and epinephrine 1:100,000 (1.7 mL)

Ultracaine DS: Articaine hydrochloride 4% and epinephrine 1:200,000 (1.7 mL) [contains sodium metabisulfite]

Ultracaine DS Forte: Articaine hydrochloride 4% and epinephrine 1:100,000 (1.7 mL) [contains sodium metabisulfite]

Administration: Adult

For submucosal infiltration and/or nerve block injection. Avoid intravascular injection. Aspirate the syringe after tissue penetration and before injection to minimize chance of direct vascular injection.

Administration: Pediatric

Dental: For submucosal infiltration and/or nerve block injection. Avoid intravascular injection. Aspirate the syringe after tissue penetration and before injection to minimize chance of direct vascular injection.

Storage/Stability

Store at 25°C (77°F) with brief excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light. Do not freeze.

Use: Labeled Indications

Dental anesthesia: Local, infiltrative, or conductive anesthesia in both simple and complex dental procedures

Metabolism/Transport Effects

Refer to individual components.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alpha1-Blockers: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Azosemide: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Benperidol: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Alpha-/Beta-Agonists and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Consider delaying skin testing until alpha/beta agonists are no longer required and their effects have diminished, or use a histamine skin test as a positive control to assess a patient’s ability to mount a wheal and flare response. *Risk D: Consider Therapy Modification*

Beta-Blockers (Beta1 Selective): May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Beta-Blockers (Nonselective): May increase hypertensive effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Beta-Blockers (with Alpha-Blocking Properties): May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Blonanserin: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Bromocriptine: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives to this combination when possible. If combined, monitor for hypertension and tachycardia, and do not coadminister these agents for more than 10 days. *Risk D: Consider Therapy Modification*

Bromperidol: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

BUPivacaine (Liposomal): Local Anesthetics may increase adverse/toxic effects of BUPivacaine (Liposomal). Management: Liposomal bupivacaine should not be administered with local anesthetics, but may be administered 20 minutes or more after lidocaine. Avoid all local anesthetics within 96 hours after administration of liposomal bupivacaine. *Risk X: Avoid*

BUPivacaine: Local Anesthetics may increase adverse/toxic effects of BUPivacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant

(Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Cardiac Glycosides: EPINEPHrine (Systemic) may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

Chloroprocaine (Systemic): May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Chlorprothixene: May increase adverse/toxic effects of EPINEPHrine (Systemic). Specifically, paradoxical hypotension and tachycardia may be increased. EPINEPHrine (Systemic) may increase therapeutic effects of Chlorprothixene. *Risk C: Monitor*

CloZAPine: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

COMT Inhibitors: May increase serum concentrations of COMT Substrates. *Risk C: Monitor*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Diuretics: May increase arrhythmogenic effects of EPINEPHrine (Systemic). Diuretics may decrease vasopressor effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May increase vasoconstricting effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Haloperidol: May decrease vasoconstricting effects of EPINEPHrine (Systemic). Management: Consider alternatives to this combination and monitor for reduced epinephrine efficacy, and possible paradoxical effects (ie, hypotension), when combined. Use of alternative vasopressor agents (eg, phenylephrine, metaraminol, norepinephrine) is preferred. *Risk D: Consider Therapy Modification*

Hexoprenaline: May increase adverse/toxic effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Hyaluronidase: May increase vasoconstricting effects of Alpha-/Beta-Agonists. Management: Do not use hyaluronidase to enhance the dispersion or absorption of alpha-/beta-agonists. Use of hyaluronidase for other purposes in patients receiving alpha-/beta-agonists may be considered as clinically indicated. *Risk D: Consider Therapy Modification*

Inhalational Anesthetics: May increase arrhythmogenic effects of EPINEPHrine (Systemic). Management: Administer epinephrine with added caution in patients receiving, or who have recently received, inhalational anesthetics. Use lower than normal doses of epinephrine and monitor for the development of cardiac arrhythmias. *Risk D: Consider Therapy Modification*

Isoproterenol: May increase therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lisuride: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives when possible, as product labeling states the concomitant use of lisuride with alpha-/beta-agonists is not recommended. If combined, monitor for increases in blood pressure or hypertensive crises. *Risk D: Consider Therapy Modification*

Local Anesthetics: Articaine may increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Melperone: May increase adverse/toxic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase hypertensive effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pergolide: May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Promethazine: May decrease vasoconstricting effects of EPINEPHrine (Systemic). Management: Avoid epinephrine and consider norepinephrine or phenylephrine when treating hypotension due to promethazine overdose. Consider alternative vasocontrictors in patients treated with promethazine. This combination may be indicated in anaphylaxis treatment. *Risk D: Consider Therapy Modification*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

ROPivacaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase tachycardic effects of Alpha-/Beta-Agonists. Serotonin/Norepinephrine Reuptake Inhibitor may increase vasopressor effects of Alpha-/Beta-Agonists. Management: If possible, avoid coadministration of direct-acting alpha-/beta-agonists and serotonin/norepinephrine reuptake inhibitors. If coadministered, monitor for increased sympathomimetic effects (eg, increased blood pressure, chest pain, headache). *Risk D: Consider Therapy Modification*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase vasopressor effects of Alpha-/Beta-Agonists. Management: Avoid, if possible, the use of alpha-/beta-agonists in patients receiving tricyclic antidepressants. If combined, monitor for evidence of increased pressor effects and consider reductions in initial dosages of the alpha-/beta-agonist. *Risk D: Consider Therapy Modification*

Vasopressin: Alpha-/Beta-Agonists (Direct-Acting) may increase hypertensive effects of Vasopressin. The effect of other hemodynamic parameters may also be enhanced. *Risk C: Monitor*

Pregnancy Considerations

Adverse events have been observed in some animal reproduction studies using this combination. Articaine and epinephrine cross the placenta (Sandler 1964; Strasser 1977).

Breastfeeding Considerations

It is not known if articaine or epinephrine are present in breast milk.

The manufacturer recommends that caution be exercised when administering articaine/epinephrine to patients who are breastfeeding; consideration may be given to pumping and discarding milk for 4 hours after the last dose. In general, women administered single dose local anesthesia for dental procedures may resume breastfeeding once they are awake and stable (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Cardiovascular and respiratory vital signs; state of consciousness after each injection; CNS toxicity.

Mechanism of Action

Articaine: Blocks both the initiation and conduction of nerve impulses by increasing the threshold for electrical excitation in the nerve, slowing the propagation of the nerve impulse, and reducing the rate of rise of the action potential.

Epinephrine: Increases the duration of action of articaine by causing vasoconstriction (via alpha effects) which slows the vascular absorption of articaine.

Pharmacokinetics (Adult Data Unless Noted)

Also refer to the Epinephrine (Systemic) monograph.

Onset of action: 1 to 9 minutes

Duration: Complete anesthesia: ~1 hour (infiltration); ~2 hours (nerve block)

Distribution: Articaine: ~60% to 80%, bound to albumin and gamma globulins

Metabolism: Articaine: Hepatic via plasma carboxyesterase to articainic acid (inactive)

Half-life elimination: Articaine/epinephrine: 43.8 to 44.4 minutes

Time to peak, plasma: Articaine: ~25 minutes (single dose); 48 minutes (3 doses)

Excretion: Urine (primarily as metabolites)

Patient Counseling Points

(For additional information see ”Articaine and epinephrine: Patient drug information”)

What is this drug used for?

• It is used before dental care to numb the area.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Injection site irritation

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Methemoglobinemia like blue or gray color of the lips, nails, or skin; abnormal heartbeat; seizures; severe dizziness or passing out; severe headache; fatigue; loss of strength and energy; or shortness of breath

• Seizure

• Trouble breathing

• Slow breathing

• Shallow breathing

• Abnormal heartbeat

• Lightheadedness

• Depression

• Chest pain

• Dizziness

• Passing out

• Restlessness

• Anxiety

• Noise or ringing in the ears

• Mouth numbness or tingling

• Blurred vision

• Tremors

• Fatigue

• Confusion

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

Find brand name(s) by country or region (for United States and Canada see separate Brand Names sections)

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Anescart forte | Septanest con adrenalina | Totalcaina forte;
- (AT) Austria: Orabloc | Septanest mit Epinephrin | Sopira citocartin;
- (AU) Australia: Septanest;
- (BE) Belgium: Septanest special | Ubistesin Adrenaline;

- (BG) Bulgaria: Dentocaine;
- (BR) Brazil: Articaine | Septanest;
- (CH) Switzerland: Alphacaine | Septanest adrenalin;
- (CI) Côte d'Ivoire: Septanest adrenalinee;
- (CO) Colombia: Artheek | Septanest con adrenalina;
- (CR) Costa Rica: Artheek | Turbocaina;
- (CZ) Czech Republic: Septanest S Adrenalinem | Ubistesin;
- (DE) Germany: Artinestol | Septanest | Sopira citocartin mit epinephrin;
- (EC) Ecuador: Artheek | Septanest con adrenalina | Turbocaina;
- (EE) Estonia: Dentocaine | Septanest | Septanest 1/100000 | Septanest 1/200000 | Ultracain D-suprarenin;
- (EG) Egypt: Alexadricaine;
- (ES) Spain: Artinibsa;
- (FI) Finland: Dentocaine | Septocaine | Ubistesin;
- (FR) France: Predesic adrenaline | Septanest adrenalinee;
- (GB) United Kingdom: Anestadent | Septanest;
- (GR) Greece: Septanest | Ubistesin;
- (HR) Croatia: Septanest | Ubistesin | Ubistesin forte;
- (HU) Hungary: Artinibsa | Septanest;
- (IT) Italy: Alfacaina N | Alfacaina sp | Articaina ogna | Artin | Cartidont;
- (JO) Jordan: Primacaine adrenaline;
- (KR) Korea, Republic of: Posicaine | Septanest;
- (LT) Lithuania: Citocartin | Ubistesin;
- (LU) Luxembourg: Dentocaine;
- (LV) Latvia: Dentocaine | Septanest | Ubistesin;
- (MA) Morocco: Artinibsa | Primacaine adrenaline | Septanest adrenalinee;
- (MX) Mexico: Medicaine | Turbocaina;
- (MY) Malaysia: Posicaine | Primacaine adrenaline | Septanest | Ubistesin;
- (NL) Netherlands: Septanest | Ultracain d-s;
- (NO) Norway: Septocaine | Septocaine forte;
- (NZ) New Zealand: Ardanest | Ubistesin;
- (PA) Panama: Artheek;
- (PL) Poland: Citocartin | Septanest z adrenalina;
- (PR) Puerto Rico: Articadent dental | Orabloc | Septocaine;
- (PT) Portugal: Artinibsa;
- (PY) Paraguay: Septanest con adrenalina;
- (QA) Qatar: Artinibsa with Epinephrine | Orabloc | Septanest;
- (RO) Romania: Artidental | Ubistesin;
- (RU) Russian Federation: Alphacaine sp | Articain | Articain df | Articain+adrenalin | Articaine df | Articaine Inibsa | Articaine+epinephrine | Artifrin | Artifrin forte | Artikain s adrenalinom | Brilocaine adrenaline | Citocartin | Orabloc | Primacaine adrenaline | Septanest with adrenaline | Ubistesin | Ubistesin forte | Ultracain d-s;
- (SE) Sweden: Septocaine | Septocaine forte | Ubistesin | Ubistesin forte;
- (SI) Slovenia: Septanestepi | Ubistesin | Ultracain d-s;
- (SK) Slovakia: Orabloc | Septanest;
- (TH) Thailand: Septanest | Septanest sp | Ubistesin | Ubistesin forte;
- (TR) Turkey: Dentacaine fort;
- (TW) Taiwan: Orabloc | Posicaine | Septanest with adrenaline | Ubistesin | Ubistesin forte;
- (UA) Ukraine: Artifrin | Brilocaine adrenaline | Orabloc | Ultracain d-s;
- (UY) Uruguay: Carticaina forte epicaris | Turbocaina;
- (ZA) South Africa: Septanest with adrenaline | Ubistesin | Ubistesin forte

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